

JEE MAIN REVISION MASTER: GRAVITATION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 208 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Gravitational Field & Potential	129 PYQs	Field & Potential
Very High	Orbital Velocity, Escape Velocity & Kepler's Laws	88 PYQs	Satellites & Orbits

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Gravitational Field & Potential	$g_h \approx g \left(1 - \frac{2h}{R}\right)$; ($h \ll R$), $\quad g_d = g \left(1 - \frac{d}{R}\right)$, $\quad V(r) = -\frac{GM}{r}$
Orbital Velocity, Escape Velocity & Kepler's Laws	$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \approx 11.2 \text{ km/s}$, $\quad v_o = \sqrt{\frac{GM}{R}}$, $\quad v_e = \sqrt{2} v_o$, $\quad T^2 = \frac{4\pi^2}{GM} r^3$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Gravitational Field & Potential (Very High Yield — 129 PYQs)

Core Principle: Newton's law of gravitation, variation of g with height, depth, and latitude λ ($g' = g - \omega^2 R \cos^2 \lambda$).

■ **Caution / JEE Pitfall:** At $h = R$ above Earth, $g_h = (g / (1 + 1)^2) = (g / 4)$. The approximation $(1 - 2h/R)$ fails when h is comparable to R !

■ **Speed Trick:** Decrease in g at height h is double the decrease at depth d for $h = d \ll R$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A body of mass m is taken from the surface of earth to a height equal to twice the radius of earth (R_e). The increase in potential energy..." [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "Net gravitational force at the center of a square is found to be F_1 when four particles having mass $M, 2M, 3M$ and $4M$ are placed at the corners..." [Ans: A]
- [2026 Morning Shift] Q#2 (MCQ): "When one moves from a point 16 km below the earth's surface to a point 16 km above the earth's surface. The change in g is approximately..." [Ans: B]

3.2 Orbital Velocity, Escape Velocity & Kepler's Laws (Very High Yield — 88 PYQs)

Core Principle: Orbital mechanics of satellites, escape speed from planetary surfaces, and Kepler's three laws of planetary motion.

■ **Caution / JEE Pitfall:** Total energy of a bound satellite is negative: $E = -K = (1/2)U = -(GMm/2r)$.

■ **Speed Trick:** Geostationary satellite period is 24 hours, orbits at height $\approx 36,000$ km strictly in the equatorial plane from West to East.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "A planet (P_1) is moving around the star of mass $2M$ in the orbit of radius R . Another planet (P_2) is moving around another..." **[Ans: B]**
- **[2026 Morning Shift] Q#2 (MCQ):** "Initially a satellite of 100 kg is in a circular orbit of radius $1.5R_E$. This satellite can be moved to a circular orbit of radius $3R_E$..." **[Ans: B]**
- **[2026 Evening Shift] Q#3 (MCQ):** "Given below are two statements : Statement I : A satellite is moving around earth in the orbit very close to the earth surface. The time per..." **[Ans: C]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Evening Shift (Q#1)

A body of mass m is taken from the surface of earth to a height equal to twice the radius of earth (R_E). The increase in potential energy will be ____ . (g is acceleration due to gravity at the surface of earth)

- (A) $(1/2)mgR_E$
- (B) $(3/4)mgR_E$
- (C) $(1/4)mgR_E$
- (D) $(2/3)mgR_E$ **[Correct]**

Correct Answer: D

Solution Analysis: The potential energy (U) of a mass m at a distance r from the centre of the Earth is: $U = -\frac{GMm}{r}$. As we know that $g = \frac{GM}{R_E^2}$, so $GM = gR_E^2$. $U = -\frac{mgR_E^2}{r}$. At the surface of the Earth, the distance from the centre is $r_1 = R_E$. So, the initial gravitational potential energy of body is; $U_i = -\frac{mgR_E^2}{R_E} = -mgR_E$. The body is taken to a height $h = 2R_E$ above the surface. The distance from the centre is $r_2 = R_E + h = R_E + 2R_E = 3R_E$; $U_f = -\frac{mgR_E^2}{3R_E} = -\frac{1}{3}mgR_E$. So, the increase in p

Benchmark Problem 2: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

A planet (P_1) is moving around the star of mass $2M$ in the orbit of radius R . Another planet (P_2) is moving around another star of mass $4M$ in a orbit of radius $2R$. Ratio of time periods of revolution of P_2 and P_1 is _____.

- (A) $(1 / 2)$
- (B) 2 **[Correct]**
- (C) 4
- (D) $(1 / 4)$

Correct Answer: B

Solution Analysis: For a planet of mass m orbiting a star of mass M_s at a distance r , the gravitational force provides the necessary centripetal force for circular motion. Centripetal Force = Gravitational Force $\frac{mv^2}{r} = \frac{GM_s m}{r^2}$ So, the orbital velocity $v : v = \sqrt{\frac{GM_s}{r}}$ The time period T is the time taken to complete one full orbit of circumference $2\pi r$: $T = (2\pi r / v) = \frac{2\pi r}{\sqrt{\frac{GM_s}{r}}} = \frac{2\pi r}{\sqrt{r}} \sqrt{GM_s} = \frac{2\pi r^{3/2}}{\sqrt{GM_s}}$ On squaring both sides $T^2 = \frac{4\pi^2 r^3}{GM_s}$ $\therefore$